1. **TABLE OF CONTENTS (TOC)**


DEFINITIONS

OVERVIEW

EXAM PREPARATION

SCENE ASSESSMENT

LIVE DEVICE EXAM

STATIC DEVICE EXAM

FORENSIC DATA ANALYSIS

REPORTING

ACADEMIC INTEGRITY

APPLICABILITY

REFERENCES

APPENDIX



UNIVERSITY AT ALBANY

State University of New York

## 2. DEFINITIONS

TOC

2.1. WRITTEN INSTRUCTIONS – directions disseminated in writing by the instructor that may be communicated through any or all the following methods:

2.1.1. *Course Requirements* documents,
2.1.2. *Module* Instructions,
2.1.3. *BrightSpace* postings,
2.1.4. *Email* message,
2.1.5. Course *Syllabus*,
2.1.6. Classroom Erase Board,
2.1.7. Any other written dissemination of instructions by the instructor.

2.2. REQUIREMENT/REQUIRED – written instruction that outlines a task and/or activity that must be completed or performed by the student.

2.3. SCOPE OF AUTHORITY: The specific and allowable tasks and activities that each student is required to perform as part of the coursework, as directed by the written instructions.

2.4. MUST: It is a requirement for a specific task or activity to be completed or performed by the student

2.5. SHALL**:** It is a requirement for a specific task or activity to be completed or performed by the student.

2.6. ENSURE: Reminder that a specific outcome or method must be achieved by the student.

2.7. RECOMMENDATION/RECOMMENDED: Information that may facilitate the proper application of the written instructions. This term is not a requirement.

2.8. NOTE: It is a recommendation or reminder to the student that may facilitate the proper application of the written instructions. This term is not a requirement.

2.9. DIGITAL EVIDENCE: may include any of the following:

2.9.1. Computers/laptops/notebooks.
2.9.2. Mobile Devices.
2.9.3. Cameras.
2.9.4. Video Recorders.
2.9.5. Hard Disk Drives.
2.9.6. Removable Media.
2.9.7. Any other device, media, peripheral, file, data, item, or EXAM DATA, whether physical or virtual, which may transmit, store, or receive electronic information of potential probative value.

2.10. EXAM DATA: All files or data collected during an examination or analysis of digital evidence.

2.11. COLLECT: Digital Evidence files or data that may be obtained, exported, copied, acquired or transferred from an evidence device or media to an external storage device or media.

2.12. CUSTODY: To possess, handle, protect, examine, analyze, or transfer digital evidence.

2.13. TOOL DATA: Any data automatically generated by tool or utility during an examination.

2.14. TOOL REPORT: Any report or file generated by the user using a tool during an examination.

UNIVERSITY AT ALBANY
State University of New York

## 3. OVERVIEW

TOC

3.1. All students must follow the written instructions, as defined in this document.

3.2. Grading and student assessments by the instructor are based on the course rubrics, which are derived from the written instructions.

3.3. Lectures and tutorials by the instructor or Peer Educators/Teaching assistants are used to support the written instructions and to be used as a guide only.

3.3.1. NOTE: Lectures and tutorials DO NOT supersede or replace the written instructions.

3.4. Course documents such as best practices, standards, guides and manuals posted via *BrightSpace* are used only for the purpose of assisting students in understanding the basis of the written instructions.

3.4.1. NOTE: Such documents do NOT replace or supersede the course written instructions or requirements.

3.5. Digital Forensic Investigations may include any or all the following tasks and activities:

3.5.1. Assessment of an incident scene (physical and/or virtual).
3.5.2. Handling of items that may be considered DIGITAL EVIDENCE.
3.5.3. Live and/or static device examination of digital evidence using forensically sound methods/tools.
3.5.4. Reporting all performed activities and tasks in required reporting.
3.5.5. Any other activity or task, as directed by the instructor.

3.6. Scene Assessment, Evidence Handling, Forensic Examination and Data Analysis must be performed based on the scope of authority of the coursework.

3.7. Digital Evidence must be handled and examined while minimizing the risk of any contamination, modification or action that may affect its original state or potential probative value.

3.8. Each student must use the following forensic methods and/or tools to minimize the risk of affecting the original state of digital evidence:

3.8.1. Use forensically sound methods, techniques and tools to perform device examinations.
3.8.2. Mounting evidence device partitions as Read-Only.
3.8.3. Collecting EXAM DATA directly to an export drive or host-computer shared folder.
3.8.4. Computing hash values of the exam data files before they are collected, and after collection.
3.8.5. Using Write-blockers for disks and media.
3.8.6. Using Faraday enclosures for mobile devices.

3.9. Once handled, examined or analyzed, all digital evidence must be kept in custody of a person, place, or thing, as well as maintaining accurate documentation that identifies each specific item description, and dates/times of their transfer or custody exchange, such as:

3.9.1. Person: name of an individual (student, teammate, teaching assistant, instructor, etc.).
3.9.2. Place: Describe the Physical location (Evidence Locker/Shelf/Room), Logical/Virtual location (*BrightSpace*, Network storage, Cloud Storage, etc.).
3.9.3. Things: physical container (box, envelope, etc.), logical/virtual container (USB drive, zip file, virtual machine, host computer shared folder, etc.).

3.10. Do not collect any file or data that exceeds the scope of authority.

3.11. Examination of a device, file or data that may include the collection of digital evidence must be conducted using the assigned Incident Response (IR) tool or forensic utilities, and other applications, as directed by the written instructions.

UNIVERSITY AT ALBANY State University of New York

3.12. Media containing the Incident Response (IR) tool and its utilities or applications shall NOT be considered as digital evidence, UNLESS said media also contains EXAM DATA that was collected during an examination.

3.13. Unless otherwise authorized by written instructions, the evidence device's native utilities, applications, programs, or executables must NOT be used to access, create, extract, or export stored files, data, or information (EXAM DATA) during examination.

3.14. EXAM DATA that is collected during an examination must NOT be saved in the examined evidence device since this may alter, modify, tamper, or destroy relevant data or lessen the probative value of said digital evidence, or exceed the scope of authority.

3.14.1. EXAM DATA must be collected directly to removable media (EXPORT drive) or Host-computer shared folder (VM).

3.15. EXAM DATA **screenshots**, **tool reports** or other exam data files that are created during an examination must be labeled or named so that it clearly identifies the *IR Tool* that was used during the examination, the *IR Tool* utility or application that displayed or created the collected information, and a general description of the contents of the file. (Refer to the EXAMPLE exam data file names below)

3.15.1. **Screenshot** files must be saved in ***png*** or ***jpg*** formats ONLY.

3.15.1.1. EXAMPLE: *kali_gparted_computerpartitions.jpg*
3.15.1.2. EXAMPLE: *knoppix_file manager_windowsusers.png*
3.15.1.3. EXAMPLE: *caine_terminal_fdisk_command.jpg*
3.15.1.4. EXAMPLE: *deft_disks_computerpartitions.png*
3.15.1.5. EXAMPLE: *paladin_devicemanager_device_partitions_filesystems.jpg*

3.15.2. **Tool reports** files must be saved in ***pdf, htm***, or ***html*** formats ONLY.

3.15.2.1. EXAMPLE: *CAINE – WinAudit – ComputerInfo.htm*
3.15.2.2. EXAMPLE: *DEFT – Autopsy – LinuxUsers.htm*
3.15.2.3. EXAMPLE: *CAINE – USBDeview – USB devices.htm*
3.15.2.4. EXAMPLE: *HashMyFiles – Exam Data – Hash Values.html*
3.15.2.5. EXAMPLE: *Dir2Html – Reporting – Documentation.html*

3.15.3. Memory or acquisition **exam data file** names.

3.15.3.1. EXAMPLE: *CAINE – Guymager – disk_acquisition_image.e01*
3.15.3.2. EXAMPLE: *Magnet Capture – memory_acquisition.raw*
3.15.3.3. EXAMPLE: *Belkasoft - MemCapture – memory acquisition.mem*
3.15.3.4. EXAMPLE: *FTK imager – memory dump.mem*
3.15.3.5. EXAMPLE: *EnCase imager – ram dump.e01*

3.16. Ensure all screenshots show all the required and viewable information without truncating or cutting off any of the available information that may be displayed in the screenshot.

3.17. Students must retain the original file name when collecting exam data files found in the evidence device that already have a specific file name.

3.17.1. **NOTE:** Do not change the original file names to minimize risk of corrupting contents of said files.

3.18. Students must retain a copy of the EXAM DATA for the purpose of completing other tasks, such as data analysis, technical reviews, reporting, submissions and any other related activity or task.

3.19. All scene assessments, device examinations, analyzes, and reporting must comply with the written instructions.

UNIVERSITY AT ALBANY State University of New York

## 4. EXAM PREPARATION

TOC

4.1. Prepare your authorized *IR Tool* and/or utilities in an external device or media (EXPORT drive) to use during the examination of digital evidence.

4.2. Ensure the IR Tool and utilities are tested and validated PRIOR to perform an examination in a case and NOT during the examination or using the evidence device.

4.3. EXPORT drive must be properly prepared and formatted prior to starting the examination so it may be used with several forensic tools and can store EXAM DATA during an examination.

4.3.1. Recommendation: EXPORT drive should be formatted FAT32, VFAT or ExFAT.

4.4. Student must label the EXPORT drive as '**export**' or '**exportdrive'** to easily identify said drive during an examination.

4.5. If using an additional IR Tool drive to use as an export drive, students must then create a folder labeled '**export**' in the root of the IR Tool drive to save the exam data collected during the examination.

4.5.1. NOTE: Avoid changing any of the IR tool folders/files to minimize risk of corrupting IR tool functions.

4.6. If using a host-computer shared folder during an examination of a virtual machine, student must create a folder labeled '**export**' to save the exam data collected during the examination and a '**tools**' folder to store the required IR Tool utilities to use during an exam of the VM.

4.6.1. **NOTE:** Do not create or copy any exam data, folders or files in an evidence computer or virtual machine.

4.7. EXAM DATA must be organized in case folders and sub-folders named in such a manner that it clearly identifies and organizes the content of each folder and sub-folder.

4.8. EXAM DATA must be organized in the following folders and sub-folders and must be divided by the IR Tool used during the examination(s):

4.8.1. **casename** folder (Example folder name: *module1_examdata_yourlastname*),

4.8.1.1. **verifications** sub-folder (collection and verification hash value files only),

4.8.1.2. **toolname** sub-folder (label with the IR tool name, such as kali, knoppix, caine, etc.),

4.8.1.2.1. **export** sub-folder (folders/files exported by the user during exam),

4.8.1.2.2. **screenshots** sub-folder (exported by user during exam),

4.8.1.2.3. **tooldata** sub-folder (folders/files automatically generated by a tool during exam),

4.8.1.2.4. **toolreports** sub-folder (reports generated by the user using a tool during exam).

4.9. Students must copy all examination EXAM DATA into the appropriate case folder and sub-folder prior to submission of the exam data for assessment.

4.9.1. Do NOT delete any folder/sub-folder, even if it is empty of exam data.

## 5. SCENE ASSESSMENT

TOC

5.1. PRIOR to handling or taking custody of any digital evidence, each student must assess the incident scene based on the case scenario and scope of authority.

5.2. Incident scene assessment must be documented, as outlined in the Reporting section of this document, and must include at least the following information:

5.2.1. Date and time of the arrival, departure and/or presence at the scene,

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5.2.2. overall description of the scene(s),
5.2.3. exact physical and virtual locations of the scene(s),
5.2.4. handling of all the evidence items at the scene(s),
5.2.5. individuals present at the scene,
5.2.6. actions or tasks performed by the student during the assessment of the scene(s),
5.2.7. any other detail associated with the scene and the evidence found at the scene.

5.3. Physical scenes being assessed by students must be documented using photographs and the appropriate worksheets, as outlined in the Reporting section, so that it clearly and accurately describes all the details associated with the scene and any potential items of evidence at the scene(s).

5.4. A minimum of 4 photographs must be obtained at the physical scene that include the following views:
5.4.1. Overall scene (room),
5.4.2. Mid-Range (study/work area),
5.4.3. Close-up (evidence item or items),
5.4.4. Computer BIOS information screens,
5.4.5. Other photos, if more than one item of evidence is found at the scene.
5.4.6. NOTE: Refer to the *Appendix* section of this document for examples of physical scene and evidence photos.

5.5. Physical scene photographs must clearly and accurately depict each item of digital evidence that is handled or examined by students, including but not limited to, the following:
5.5.1. Make, model, serial number of every evidence item,
5.5.2. EXPORT drive containing evidence exam data collected during examination,
5.5.3. Any other details about the physical scene, as well as the evidence items at the scene.
5.5.4. Any other relevant or unique device information.
5.5.5. Computer BIOS or FIRMWARE information (static exam of a physical computer only),

5.6. At a minimum, the following information must be documented with photographs from the device's BIOS or firmware when prior to performing a static device examination of a physical computer/laptop.
5.6.1. BIOS Date/Time,
5.6.2. Hard Disk/Virtual Drive (HDD) information (make, model, serial number, size),
5.6.3. Random Access Memory size,
5.6.4. Boot Sequence/Order,
5.6.5. Virtual Machine (VM) or other application that is running within an evidence host computer and visible on the screen or desktop of the host computer.

5.7. All incident scene and evidence related photographs taken during the scene assessment shall be deemed as **REPORTING** and must NOT be submitted with the evidence EXAM DATA.

5.8. Refer to the **REPORTING** section for additional instructions relevant to Reporting requirements.

## 6. LIVE DEVICE EXAM

TOC

6.1. Examination must be conducted ONLY while the device's power is ON and the live device was already running its native operating system, thereby providing a unique opportunity to examine the contents of a device that was found ON at the scene and may be accessed via user authentication.

6.2. Live device must NOT be re-booted or shut down during the live examination as you may lose volatile information that may be relevant to the investigation, such as memory, executables, and other live processes.

UNIVERSITY AT ALBANY State University of New York

6.2.1. NOTE: Shutting down or re-booting a device that is ON at the scene may lead to an encrypted or password protected device without being able to access its data contents.

6.3. Insert the *IR Tool* and EXPORT drive into the evidence device to begin the live device examination.
6.3.1. NOTE**:** You may use an EXPORT drive that also contains the *IR Tool* utilities.
6.3.2. NOTE: You may use a share-folder in the host computer when examining a virtual machine.

6.4. After inserting the drives, collect a **screenshot** showing the date/time and time-zone of the evidence computer.
6.4.1.1. DO NOT make any changes to the evidence device or VM date/time.

6.5. Each individual EXAM DATA file must be verified and authenticated by means of a hash value PRIOR to collecting said file(s) from the evidence computer/VM to the EXPORT drive or VM shared-folder.
6.5.1. When possible, a single file must be created containing the hash values for every EXAM DATA file and said file must be stored in the root of the EXAM DATA case folder.
6.5.1.1. File name: **HashingToolName – Live Exam Data – Collection Hash Values.html**
6.5.1.2. Example: *HashMyFiles – Live Device Exam Data – Collection Hash Values.html*
6.5.2. If necessary, a screenshot showing the hash values for every EXAM DATA file may be collected and stored in the root of the EXAM DATA case folder, if a hashing tool cannot generate hash values tool report.
6.5.2.1. File name: **HashingToolName – Live Exam Data – Collection Hash Values.png**
6.5.2.2. Example: *HashMyFiles – Windows Event Logs directory – Collection Hash Values.png.*

6.6. Unless otherwise directed by written instructions, EXAM DATA that must be collected includes the following files, tool reports, screenshots and ALL available information relevant from the evidence computer and/or virtual machine.

6.6.1. **Screenshots**
6.6.1.1. Show date, time and time zone of the evidence computer/VM, per section *6.4*,
6.6.1.2. Show any error messages that may be encountered during the examination,
6.6.1.3. Show relevant information when a tool report could not be generated by a specific utility,
6.6.1.3.1. Screenshot cannot be used in lieu of the tool-generated report, unless there is an error message that shows why tool report was not generated.
6.6.1.4. Show collected exam data of the EXPORT drive or host computer VM shared-folder, after completion of the examination.

6.6.2. **Tool Reports**
6.6.2.1. Recommended: Directory and file listing of the evidence partition(s).
6.6.2.2. Recommended: Hash values of all exam data files PRIOR to collection or exporting.
6.6.2.3. Windows Operating System Information (version, product name, owner, date),
6.6.2.4. Windows Computer Name,
6.6.2.5. Windows Disk/ Drives/Volumes/Media information (make, model, serial #, description),
6.6.2.6. Windows partition and file system (sizes, path, mount point, label, description, etc.),
6.6.2.7. Windows Groups and Users,
6.6.2.8. Windows System and User login history,
6.6.2.9. IP and MAC addresses, adapters and Network routing information.

6.6.3. **Files**
6.6.3.1. If individual files in the Windows partition are within the scope of the examination, then collect said exam data files, if feasible.
6.6.3.2. If individual files located in another partition accessible via the Windows OS, then collect said exam data files, if feasible.

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6.6.4. **Miscellaneous exam data**
6.6.4.1. Acquisition tool data.
6.6.4.2. Screenshot that shows any error messages that may be encountered during the examination.
6.6.4.3. Screenshot that shows the relevant information when a tool report utility was not generated or a terminal command or application did not produce the desired results.
6.6.4.4. Screenshot that shows the collected exam data in the EXPORT drive or host computer VM shared-folder, after completion of the examination.
6.6.4.5. Any other exam data (screenshot/file/tool report) that may provide basic and relevant information about the examined device, file systems, hardware, software, users, activity, contents, applications, and other probative or unique information, within the scope of authority.

6.7. After completion of the examination, it is highly recommended that students eject the export drive and insert into the personal or classroom computer to verify all required exam data has been saved to the export drive and files are NOT empty (zero bytes).
6.7.1. **NOTE**: If using VM shared-folder, verify that all required exam data was collected.

6.8. Each individual EXAM DATA file must be verified and authenticated by means of a hash value AFTER the completion of the examination and PRIOR to the submission of EXAM DATA to the instructor
6.8.1. **DO NOT** perform this hashing while still connected to the evidence computer – instead use a different computer and tool to perform this verification hashing.
6.8.2. A single file must be created containing the hash values for every EXAM DATA file and said file must be stored in the root of the EXAM DATA case folder.
6.8.3. File name: **HashingToolName – Live Device Exam Data - Verification Hash Values.html**
6.8.3.1. Example: *HashMyFiles – Live Device Exam Data - Verification Hash Values.htm*
6.8.3.2. Example: *FTK Imager – Live Device Exam Data – Verification Hash Values.htm*
6.8.4. **NOTE:** Creation of the verification hash values does not eliminate any other requirements outlined in this document.

6.9. *EXAM DATA* must be indexed using *DIR2HTML* or other utility that creates web links (htm/html) for every folder and file and said web links must work from a single web page.
6.9.1. Ensure you select indexing to be recursive, and includes file name, size and date/time information.
6.9.2. The primary DIR2HTML index file must be saved to the root of the **casename** exam data folder to ensure it properly works and has indexed and linked every folder, sub-folder and file being submitted.

6.10. Once the primary DIR2HTML index file is created, ensure it properly works and has indexed and linked every folder, sub-folder and file being submitted.
6.10.1. Do NOT move the DIR2HTML index files to a different folder/sub-folder since it will unlink all the previously linked exam data files.
6.10.2. **NOTE**: DIR2HTML indexing/linking cannot be verified after exam data has been compressed into a zip file therefore test indexing/linking PRIOR to zipping exam data.

6.11. When creating the submission zip file just compress the case folder containing all sub-folders and files and name the zip file appropriately, or as directed by the written instructions.
6.11.1. Example zip file name: *Module 1 – Live Device Exam Data – YourLastName.zip*

6.12. All tasks and activities performed by the student during the examination must be documented in detail as outlined in the **REPORTING** section and the written instructions.



UNIVERSITY AT ALBANY State University of New York

6.12.1. Include in your reporting any errors displayed or encountered during the examination.

## 7. STATIC DEVICE EXAM

TOC

7.1. The device's power must be OFF (static) prior to the commencement of the examination.

7.2. Commence the static examination of the device by using the appropriate Incident Response (IR) Tool to boot the evidence computer or virtual machine and utilize the IR Tool's utilities to collect EXAM DATA directly to the EXPORT drive, or host computer shared-folder.

7.3. After booting to the IR Tool's environment, collect a **screenshot** showing the date/time and time-zone of the evidence computer, as shown via the IR Tool.

7.4. Adjust the date, time and time zone within the IR Tool environment to reflect the examination's current date, time and time zone.

7.4.1. Utilize either the IR Tool graphic user interface (GUI) application or terminal commands to adjust settings.

7.4.2. After adjustment, collect a **screenshot** showing said current exam date, time and time zone and include screenshot in exam data submission.

7.5. Each individual EXAM DATA file must be verified and authenticated by means of a hash value PRIOR to collecting said file from the evidence computer/VM.

7.5.1. When possible, a single file must be created containing the hash values for every EXAM DATA file and said file must be stored in the root of the EXAM DATA case folder.

7.5.1.1. Example File name: **HashingToolName – Static Device Exam Data – Collection Hash Values.html**

7.5.1.2. Example: *Quickhash – Linux Logs directory – Collection Hash Values.jpg or png*

7.6. Ensure all tool reports and screenshots show all the required and available information without truncating or cutting off relevant information.

7.7. Unless otherwise directed by written instructions, EXAM DATA that must be collected includes the following files, tool reports, screenshots and ALL available information relevant from the evidence computer and/or virtual machine.

7.7.1. **Screenshots**

7.7.1.1. Date-time and Time-Zone PRIOR to starting examination, per section *7.4,*

7.7.1.2. Date-time and Time-Zone after adjustment, per section *7.5,*

7.7.1.3. Hard Disk/Virtual Drive evidence device information (make, model, serial number),

7.7.1.4. Partitions and File systems of evidence HDD (sizes, paths, mount points, labels, etc.),

7.7.1.5. Terminal commands or applications that show how evidence partitions were mounted,

7.7.1.6. Primary Windows users directory showing the users' folders,

7.7.1.7. Primary Windows Registry directory showing the registry files,

7.7.1.8. Primary Windows Event Logs directory showing the log files,

7.7.1.9. Primary directory of the Linux users' folders (/home),

7.7.1.10. Primary Linux logs directory (/var/log) showing the log files,

7.7.1.11. Primary configuration files directory (/etc) showing the configuration files,

7.7.2. **Files**

7.7.2.1. **Windows User Registry Files** (/Users)**:** *NTUSER.DAT* for each unique Windows user

7.7.2.1.1. NOTE: Do NOT include the default, public, or allusers files.

UNIVERSITY AT ALBANY State University of New York

7.7.2.2. **Windows Registry Files** (/Windows/System32/config)**:** *SAM, SECURITY, SOFTWARE, SYSTEM*.
7.7.2.2.1. NOTE: Do NOT include empty registry files.
7.7.2.3. **Windows Event Logs** (Windows/System32/winevt/Logs)**:** ALL logs (*.evt* and *.evtx* files).
7.7.2.4. **Linux User Folders** (/home): ALL user folders.
7.7.2.5. **Linux Logs** (/var/log)**:** ALL logs.
7.7.2.6. **Linux Configuration Files** (/etc): the *hostname, issue, lsb-release, networks, os-release and timezone* files.

7.7.3. **Miscellaneous exam data**
7.7.3.1. Acquisition tool data (image, text and log files).
7.7.3.2. PALADIN tool data (audit logs).
7.7.3.3. Screenshot that shows any error messages that may be encountered during the examination.
7.7.3.4. Screenshot that shows the relevant information when a tool report utility was not generated or a terminal command or application did not produce the desired results.
7.7.3.5. Screenshot that shows the collected exam data in the EXPORT drive or host computer VM shared-folder, after completion of the examination.
7.7.3.6. Any other exam data (screenshot/file/tool report) that may provide basic and relevant information about the examined device, file systems, hardware, software, users, activity, contents, applications, and other probative or unique information, within the scope of authority.

7.8. After completion of the examination and prior to shutting down the IR Tool, it is highly recommended that student unmounts the export drive and insert into the personal or classroom computer to verity all required exam data has been saved to the export drive and files are NOT empty (zero bytes).

7.9. Each individual EXAM DATA file must be verified and authenticated by means of a hash value AFTER the completion of the examination and PRIOR to the submission of EXAM DATA to the instructor;
7.9.1. **DO NOT** perform this hashing while still connected to the evidence computer – instead use a different computer and tool to perform this verification hashing.
7.9.2. A single file must be created containing the hash values for every EXAM DATA file and said file must be stored in the root of the EXAM DATA case folder.
7.9.3. File name: **HashingToolName – Static Device Exam Data - Verification Hash Values.html**
7.9.3.1. Example: *HashMyFiles – Static Device Exam Data - Verification Hash Values.html*
7.9.3.2. *Example: FTK Imager – Static Device Exam Data – Verification Hash Values.htm*
7.9.4. **NOTE:** Creation of the verification hash values does not eliminate any other requirements outlined in this document.

7.10. *EXAM DATA* must be indexed using *DIR2HTML* or other utility that creates web links (htm/html) for every folder and file, and said web links must work from a single web page;
7.10.1. Ensure you select indexing to be recursive, and includes file name, size and date/time information.
7.10.2. The primary DIR2HTML index file must be saved to the root of the **case name** folder to ensure it properly works and has indexed and linked every folder, sub-folder and file being submitted.

7.11. Once the primary DIR2HTML index file is created ensure it properly works and has indexed and linked every folder, sub-folder and file being submitted.
7.11.1. Do not move the DIR2HTML index files to a different folder/sub-folder since it will unlink all the previously linked exam data files.
7.11.2. **NOTE:** DIR2HTML indexing/linking cannot be verified after exam data has been compressed into a zip file therefore test indexing/linking PRIOR to zipping exam data.



UNIVERSITY AT ALBANY State University of New York

7.12. Exam data derived from the static device exam, hashing and dir2html indexing files must be submitted as directed by written instructions.

7.13. When creating the submission zip file just compress the case folder containing all sub-folders and files and name the zip file appropriately, or as directed by instructor via written instructions.
7.13.1. Example submission zip file name: *Module 1 – Static Device Exam Data – YourLastName.zip*

7.14. All tasks and activities performed by the student during the examination must be documented in detail as outlined in the **REPORTING** section and the written instructions.
7.14.1. Include in your reporting any errors displayed or encountered during the examination.

## 8. FORENSIC DATA ANALYSIS

TOC

8.1. Forensic Data Analysis may include the following tasks and activities:
8.1.1. Analysis of exam data while at the scene.
8.1.2. Analysis of exam data collected at the scene but analyzed at a different location.
8.1.3. An analysis of acquisition image-file and collected exam data, data fragments, artifacts or other digital evidence.
8.1.4. Utilizing specialized tools and methods for data analysis.

8.2. Students must identify information while conducting a preliminary analysis of the digital evidence to facilitate a more detailed or focused analysis, which may assist you in arriving at competent and professional findings, opinions, and conclusions.

8.3. EXAM DATA must be organized in the following case folders and must be divided by the IR Tool used during analysis:
8.3.1. **Case Name** folder (named the same as the submission file name)
8.3.1.1. **Verifications** sub-folder (verification hash values),
8.3.1.2. **Tool Name** sub-folder (folder with the name of the tool that was utilized).
8.3.1.2.1. **Tool Data** sub-folder (all folders/files automatically generated by tool),
8.3.1.2.2. **Tool Reports** sub-folder (reports generated by the student),
8.3.1.2.3. **Screenshots** sub-folder (screenshots where a tool report cannot be generated),
8.3.1.2.4. **Export** sub-folder (any other exported file or data).

8.4. Students must copy analysis EXAM DATA into the appropriate case folder and sub-folder.
8.4.1. Do NOT delete any folder/sub-folder, even if empty of exam data.

8.5. Most data analysis tools automatically generate **tool data**.
8.5.1. **Tool data** must be saved in the appropriate EXAM DATA folder.

8.6. Analysis tool-generated reports must be generated by the student during the analysis containing individual bookmarks or tags to identify specific relevant information, referred to as **tool reports**.
8.6.1. All **tool reports** must be saved in the appropriate EXAM DATA folder.

8.7. Regardless of evidence image file or exam data being analyzed, the following **tool reports** must be generated by the student PRIOR to the commencement of data analysis:
8.7.1. *Directory and File Listing of all folders and files found in the image file or exam data being analyzed,*
8.7.2. *Hash Values of all files in the image file or exam data being analyzed.*
8.7.3. **NOTE:** The above **tool reports** must be included in the analysis exam data prior to submission.

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8.8. When necessary, students may export specific files from an acquisition file or other digital evidence for further analysis and such files must be copied to the appropriate EXAM DATA folder.

8.8.1. Any file or data exported from an acquisition file or digital evidence is considered as **tool data** and must be saved in the appropriate EXAM DATA folder.

8.9. Regardless of the operating system associated with the image file or exam data, it is recommended that the following information is reviewed and analyzed during forensic data analysis to determine relevancy or value:

8.9.1. Review user folders and other user-created directories.
8.9.2. Review documents, spreadsheets, and other readable files.
8.9.3. Review images, videos and multimedia files.
8.9.4. Review emails and other communications.
8.9.5. Review Windows Registry files.
8.9.6. Review log files.
8.9.7. Review executables and other system files.
8.9.8. Review unusually named folders and files.
8.9.9. Develop keywords or other unique terms to perform relevant searches.
8.9.10. Data carving to recover and review deleted files or fragmented data found in unallocated space.
8.9.11. Use external applications or viewers to review the contents of unique files or data.

8.10. Students must bookmark possible relevant data or files identified during the data analysis to ensure information is collected and documented in the analysis tool-report.

8.10.1. Each bookmark must include a comment indicating the significance of each bookmark, including thumbnail images of any bookmarked multimedia file.

8.10.2. Upon completion of the data analysis each student a tool report is generated and collected in the appropriate exam data sub-folder with a proper file name that identifies its origin and content.

8.10.2.1. Example: *FTK – Data Analysis Report – YourLastName.pdf or htm*
8.10.2.2. Example: *Axiom – Data Analysis Report – YourLastName pdf or html*.

8.11. Student must perform *Registry Analysis* of the Windows and User Registry files found in the evidence image file or exam data using the assigned data analysis tool to identify and bookmark/tag the <u>relevant information</u>.

8.11.1. **NOTE**: At a minimum, the following Registry files must be analyzed if present in the evidence image file or exam data:

8.11.1.1. SAM, SOFTWARE, SYSTEM
8.11.1.2. NTUSER.DAT for each unique Windows user (not the default, public, allusers files).

8.12. Analysis of <u>Windows</u> Registry Files may provide potentially <u>relevant information</u> about the device, item, users, activity, contents, or other unique information, such as:

8.12.1.1. Windows product name and version.
8.12.1.2. Installation date.
8.12.1.3. Registered Owner.
8.12.1.4. Registered Organization.
8.12.1.5. Time Zone.
8.12.1.6. Computer Name.
8.12.1.7. Physical drives, Virtual drives, USB drives, Volume labels.
8.12.1.8. Usernames, Accounts, Profiles.
8.12.1.9. Login-Logoff and Computer-VM-User activity.
8.12.1.10. Network adapters, IP addresses and Mac Addresses.
8.12.1.11. Any other information relevant to the <u>scope of authority</u>.

UNIVERSITY AT ALBANY State University of New York

8.13. Student must utilize the assigned data analysis tool to generate an individual report containing the results of the registry analysis for each registry file that is analyzed.
8.13.1. Each tool report must clearly bookmark or tag the relevant information or results.
8.13.2. All Registry data analysis **tool reports** files shall be considered as EXAM DATA.

8.14. Students must perform log analysis of the collected logs and configuration files, as well as generating **tool reports** showing the findings and results of the analyses.
8.14.1. Each tool report must clearly be bookmarked or tagged the relevant information or results.
8.14.2. When log analysis tool report cannot be exported, then a screenshot may be collected showing all the bookmarks/tags of all the relevant information.
8.14.3. All log analysis **tool reports** and screenshots shall be considered as EXAM DATA and must be saved in the appropriate EXAM DATA folder.

8.15. Each individual EXAM DATA file must be verified and authenticated by means of a hash value AFTER the completion of the examination and PRIOR to the submission of EXAM DATA to the instructor.
8.15.1. A single file must be created containing the hash values for every EXAM DATA file and said file must be stored in the root of the EXAM DATA case folder.
8.15.2. File name: **HashingToolName – Registry Analysis Exam Data - Verification Hash Values. txt** or **html** or **pdf**
8.15.2.1. Example: *FTK Imager – Data Analysis Exam Data - Verification Hash Values.html*
8.15.2.2. Example: *HashMyFiles – Data Analysis Exam Data – Verification Hash Values.htm*

8.16. EXAM DATA must be indexed using *DIR2HTML* or other utility that creates web links (htm/html) for every folder and file, and said web links must work from a single web page;
8.16.1. Ensure you select indexing to be recursive, and includes file name, size and date/time information.
8.16.2. The primary *DIR2HTML* index file must be saved to the root of the **case name** folder to ensure it properly works and has indexed and linked every folder, sub-folder and file being submitted.

8.17. Once the primary *DIR2HTML* index file is created, ensure it properly works and has indexed and linked every folder, sub-folder and file being submitted.
8.17.1. Do not move the *DIR2HTML* index files to a different folder/sub-folder since it will unlink all the previously linked exam data files.
8.17.2. **NOTE:** *DIR2HTML* indexing/linking cannot be verified after exam data has been compressed into a zip file therefore test indexing/linking prior to zipping exam data.

8.18. Exam data derived from forensic data analysis, hashing and *DIR2HTML* indexing files must be included in the appropriate exam data folder(s) and in the Exam Data submission zip file.

8.19. When creating the submission zip file just compress the case folder containing all sub-folders and files and name the zip file appropriately, or as directed by instructor via written instructions.
8.19.1. Example submission zip file name: *Module 1 – Data Analysis Exam Data – YourLastName.zip*

8.20. All tasks and activities performed by the student during data analysis must be documented in detail as outlined in the **REPORTING** section and the written instructions.
8.20.1. Include in your reporting any errors displayed or encountered during data analysis.

## 9. REPORTING

TOC

9.1. Digital Forensic Investigation *Reporting* may include a scene assessment, evidence handling, device examination and/or data analysis that is documented using the appropriate worksheets, photographs



UNIVERSITY AT ALBANY State University of New York

and presentation content that accurately reports all required tasks, activities, descriptions, and information.

9.2. The **scene assessment and device examination** tasks, activities and findings must be reported in the following worksheets and files:
- 9.2.1. *DFIRLAB-01 Examiner Notes,*
- 9.2.2. *DFIRLAB-02 Incident Scene,*
- 9.2.3. *DFIRLAB-03 Chain of Custody,*
- 9.2.4. *DFIRLAB-04 Examination Report,*
- 9.2.5. *DFIRLAB-05 Computer Examination*,
- 9.2.6. *DFIRLAB-06 Hard Disk Drive Examination*, when examining the contents of a disk/drive.
- 9.2.7. *DFIRLAB-07 Removable Media Examination,* when examining the contents of external media.
- 9.2.8. *Photographs* that depict the physical incident scene and physical evidence found at the scene that must be saved in a single PDF file.
  - 9.2.8.1. Each photograph must include a description of the contents for each photo.
  - 9.2.8.2. **NOTE:** Do not submit individual photo files instead of the required PDF files.

9.3. Required REPORTING files associated with a scene assessment and device examination must be organized in the following folders/sub-folders:
- 9.3.1. **Case Name** folder (example: *Module 1 – Static Exam - YourLastName*)
  - 9.3.1.1. **Worksheets** sub-folder (all completed worksheets),
  - 9.3.1.2. **Scene Photos** sub-folder (scene/BIOS/evidence assessment photos and a description for each photo in a single PDF file),
    - 9.3.1.2.1. **NOTE:** Do not submit individual photo files.

9.4. Students must copy all required REPORTING documentation files into the appropriate case folder and sub-folder.
- 9.4.1. Do NOT delete any folder/sub-folder, even if empty of documentation files.

9.5. The **data Analysis** tasks, activities and findings must be reported with the following worksheets:
- 9.5.1. *DFIRLAB-01 Examiner Notes Worksheet*, AND
- 9.5.2. *DFIRLAB-03 Chain of Custody Worksheet*, AND
- 9.5.3. *DFIRLAB-04 Examination Report.*
- 9.5.4. **NOTE:** No photographs are required since most data analyzes occur in a controlled environment, such as a laboratory.
- 9.5.5. **NOTE:** If data analysis occurs at the scene, then scene assessment and device examination reporting requirements apply.

9.6. Utilize the *DFIRLAB-303 Worksheet Completion Instructions* document and EXAMPLES of completed DFIRLAB Worksheets to properly guide you in completing the required *DFIRLAB* worksheets.

9.7. All ***DFIRLAB* worksheets** have been provided by the instructor in editable PDF format, and must not be converted to any other type, format, or structure.

9.8. Upon completion, each ***DFIRLAB* worksheet** must be **eSigned** with an ***Adobe digital ID*** and must be locked Read-only to prevent changes to the worksheet after its completion.
- 9.8.1. Verify this process has been properly performed prior to submission.

9.9. *REPORTING* folders and files must be indexed using *DIR2HTML* or other utility that creates web links (htm/html) for every folder and file, and said web links must work from a single web page;
- 9.9.1. Ensure you select indexing to be recursive, and includes file name, size and date/time information.

UNIVERSITY AT ALBANY State University of New York

9.9.2. The primary *DIR2HTML* index file must be saved to the root of the **case name** folder to ensure it properly works and has indexed and linked every folder, sub-folder and file being submitted.

9.10. Once the primary *DIR2HTML* index file is created, ensure it properly works and has indexed and linked every folder, sub-folder and file being submitted.

9.10.1. Do not move the *DIR2HTML* index files to a different folder/sub-folder since it will unlink all the previously linked exam data files.

9.11. NOTE**:** *DIR2HTML* indexing/linking cannot be verified after exam data has been compressed into a zip file therefore test indexing/linking prior to zipping exam data.

9.12. When creating the submission zip file just compress the case folder containing all sub-folders and files and name the zip file appropriately, or as directed by instructor via written instructions.

9.12.1. Example submission zip file name: *Exercise 1 – Reporting – YourLastName.zip*

9.13. Reporting that may require a student to provide a presentation (live or recorded) to provide information about all the tasks, activities and findings associated with a specific coursework assignment and/or were previously reported in worksheets, photographs and other required documents.

9.14. Presentations must include the following preparation, format, structure, and content requirements:

9.14.1. Preparation

9.14.1.1. Ensure you prepare in a competent manner by reviewing all written instructions,
9.14.1.2. Review all reporting worksheets, photographs and documents,
9.14.1.3. Create an outline or notes to use as a guide,
9.14.1.4. It is recommended student records a practice presentation to review and adjust prior to final presentation.

9.14.2. Format

9.14.2.1. Ensure your face is shown throughout the presentation,
9.14.2.2. Recorded videos must be saved as a MP4 file,
9.14.2.3. MP4 file must be submitted instead of link to the file.

9.14.3. Structure

9.14.3.1. Start by introducing yourself,
9.14.3.2. Provide an overview of what you intend to present or discuss,
9.14.3.3. Address all the required topics as outlined in the written instructions,
9.14.3.4. Provide multimedia content that supports your presentation topics,
9.14.3.5. End by providing conclusions, findings and opinions.

9.14.4. Content

9.14.4.1. Target audience is non-technical, therefore avoid technical jargon without satisfactory definitions,
9.14.4.2. Ensure length of presentation addresses the written instructions,
9.14.4.3. Do not read your presentation – use notes or multimedia content to remind of speaking topics,
9.14.4.4. Do not use lengthy bullets or many bullets (over 4) in each slide or multimedia content,
9.14.4.5. Highlight or point to specific areas about when using complex multimedia content.

9.14.5. **NOTE:** Presentation requirements may be added, modified or deleted, per specific written instructions.

## 10. ACADEMIC INTEGRITY

TOC

10.1. As prescribed by UAlbany's *Standards of Academic Integrity*, each student must engage in the individual completion and submission of coursework for grading, as directed in written instructions.

UNIVERSITY AT ALBANY State University of New York

10.2. Copying, duplicating, fabricating, or submitting another student's coursework or content is NOT permitted and may be considered as plagiarism.

10.3. Copying, duplicating, fabricating, or submitting any coursework or content derived from any artificial intelligence algorithms, applications, or any program intended to replace or supplement the individual effort of the student is NOT permitted without explicit instructor's written permission or it may be considered as plagiarism.

## 11. APPLICABILITY

TOC

11.1. Instructor may modify any, or all course requirements and instructions to improve the efficiency and effectiveness of the class and student's coursework.

11.2. Instructor's modification of any section of this document does not invalidate this document or any other course requirement or instruction document.

## 12. REFERENCES

TOC

12.1. The document was developed using best practices and standards that were used as a guide to address the unique academic environment and associated coursework for students, as follows:

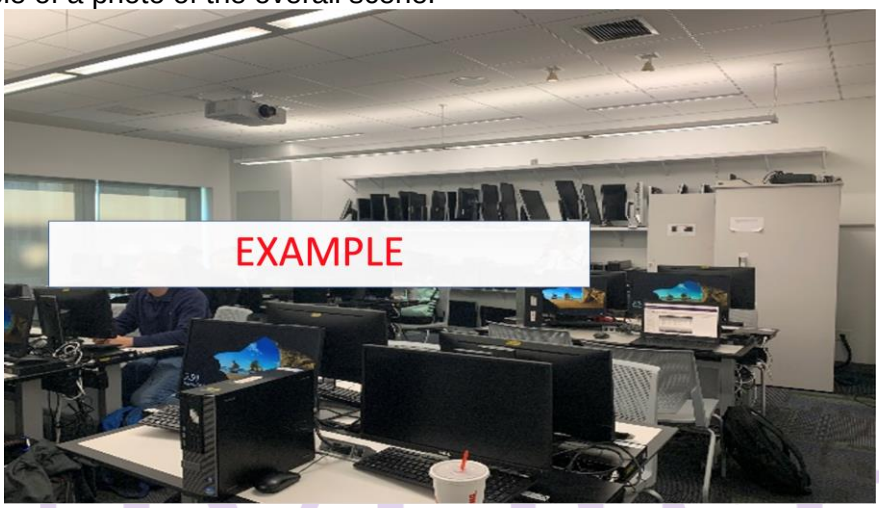
12.1.1. National Institute of Justice (NIJ).
12.1.2. Scientific Working Group on Digital Evidence (SWGDE).
12.1.3. National Institute of Standards and Technology (NIST).

UNIVERSITY AT ALBANY
State University of New York

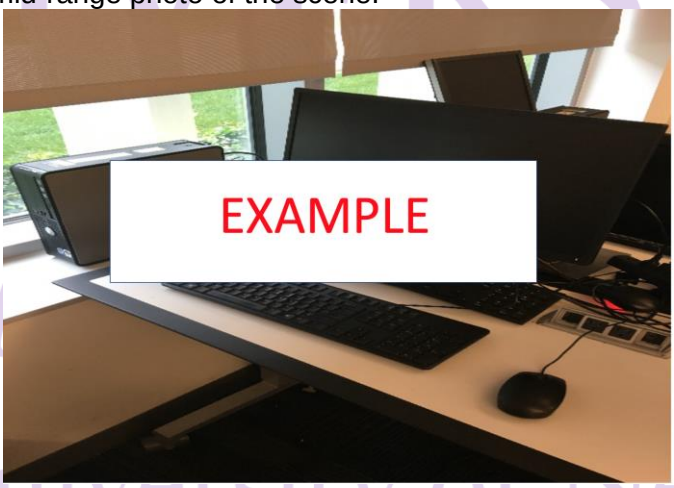
## 13. APPENDIX

TOC

13.1. Example of a photo of the overall scene.




13.2. Example of a mid-range photo of the scene.



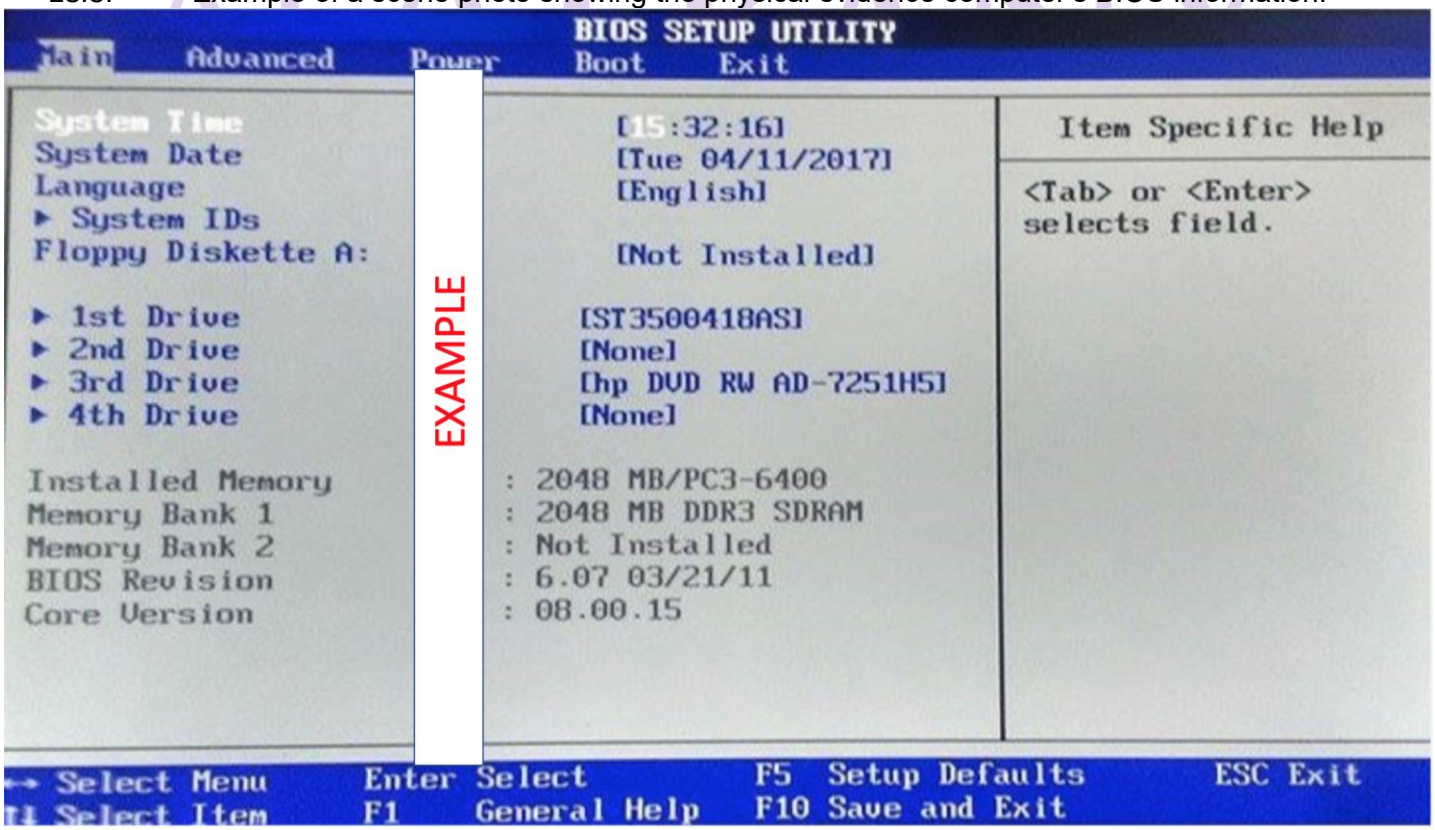

13.3. Example of a close-up photo of the evidence computer at the scene.

UNIVERSITY AT ALBANY
State University of New York

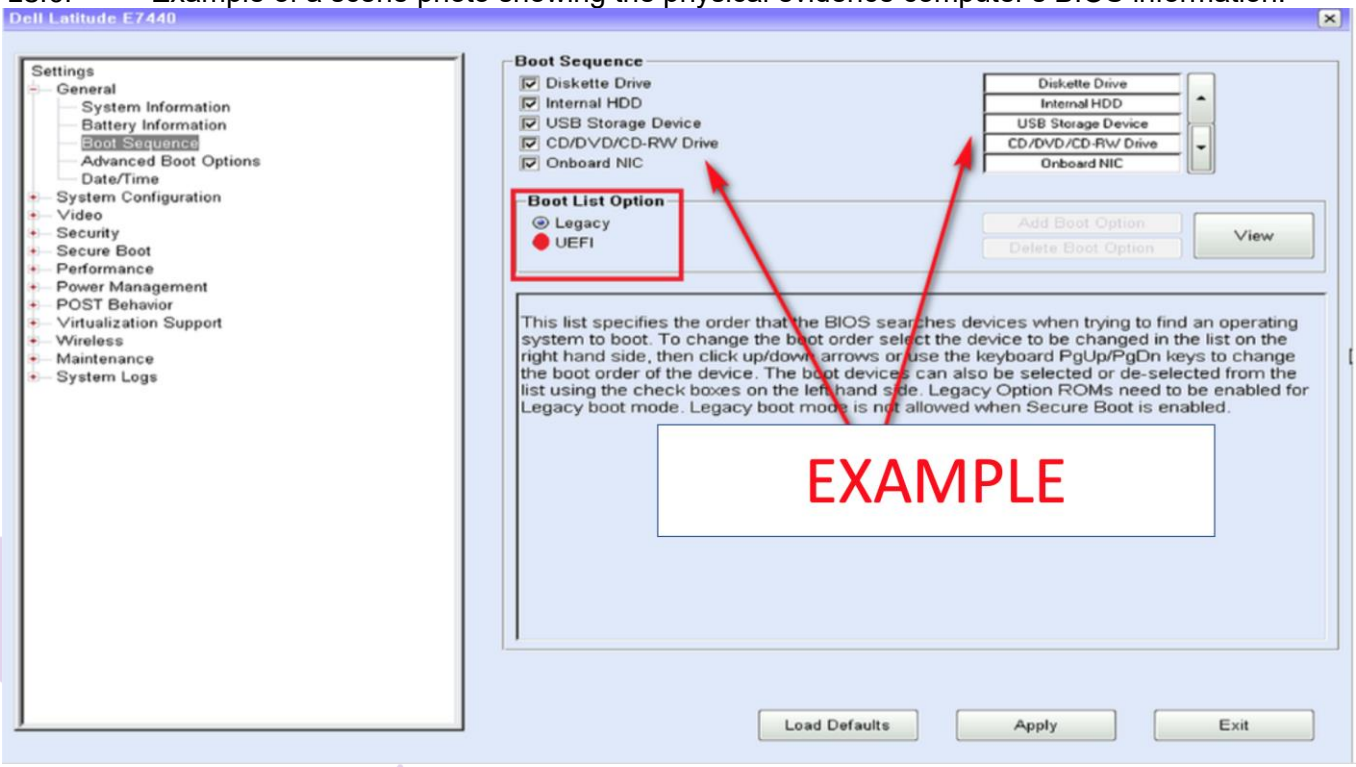
13.4. Example of a photo of the Export USB drive used to collect EXAM DATA.

UNIVERSITY AT ALBANY

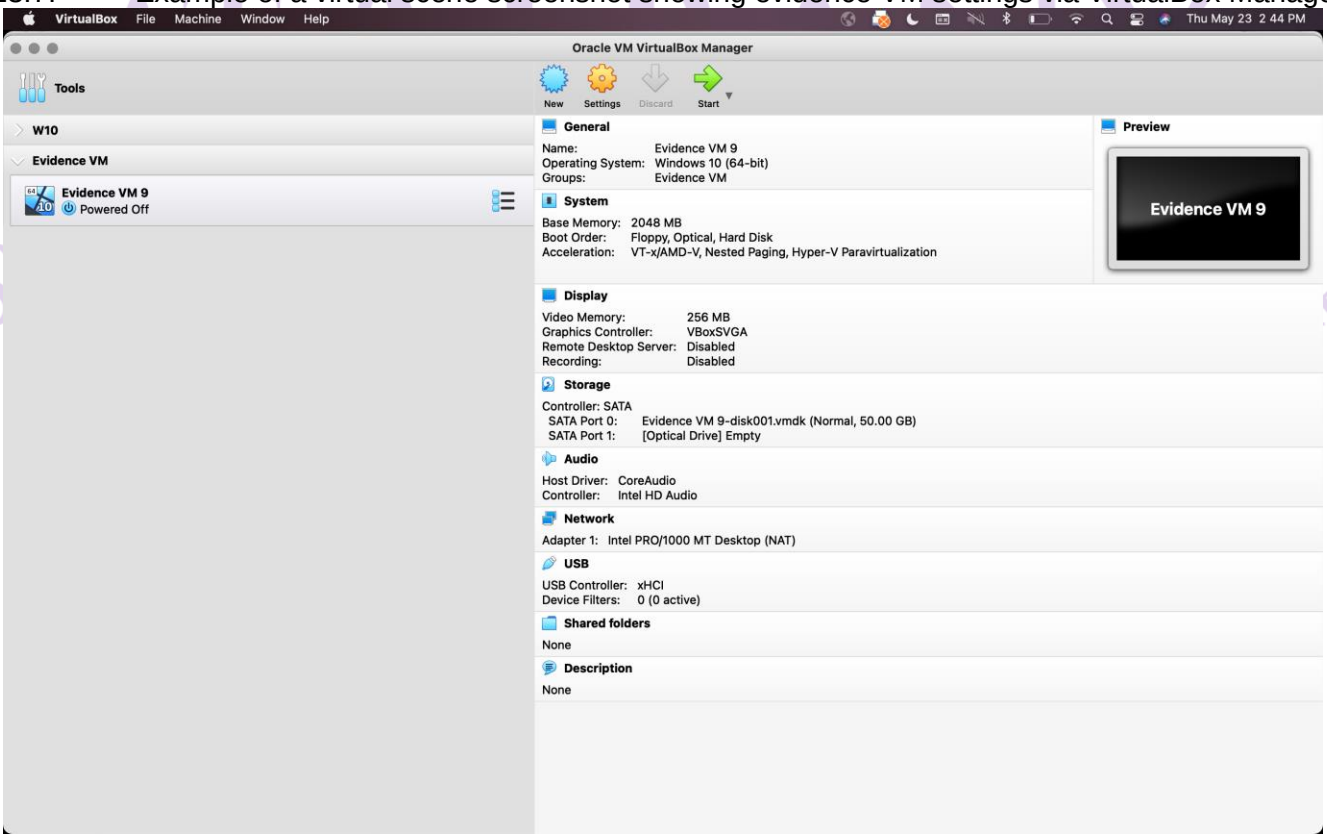
13.5. Example of a scene photo showing the physical evidence computer's BIOS information.



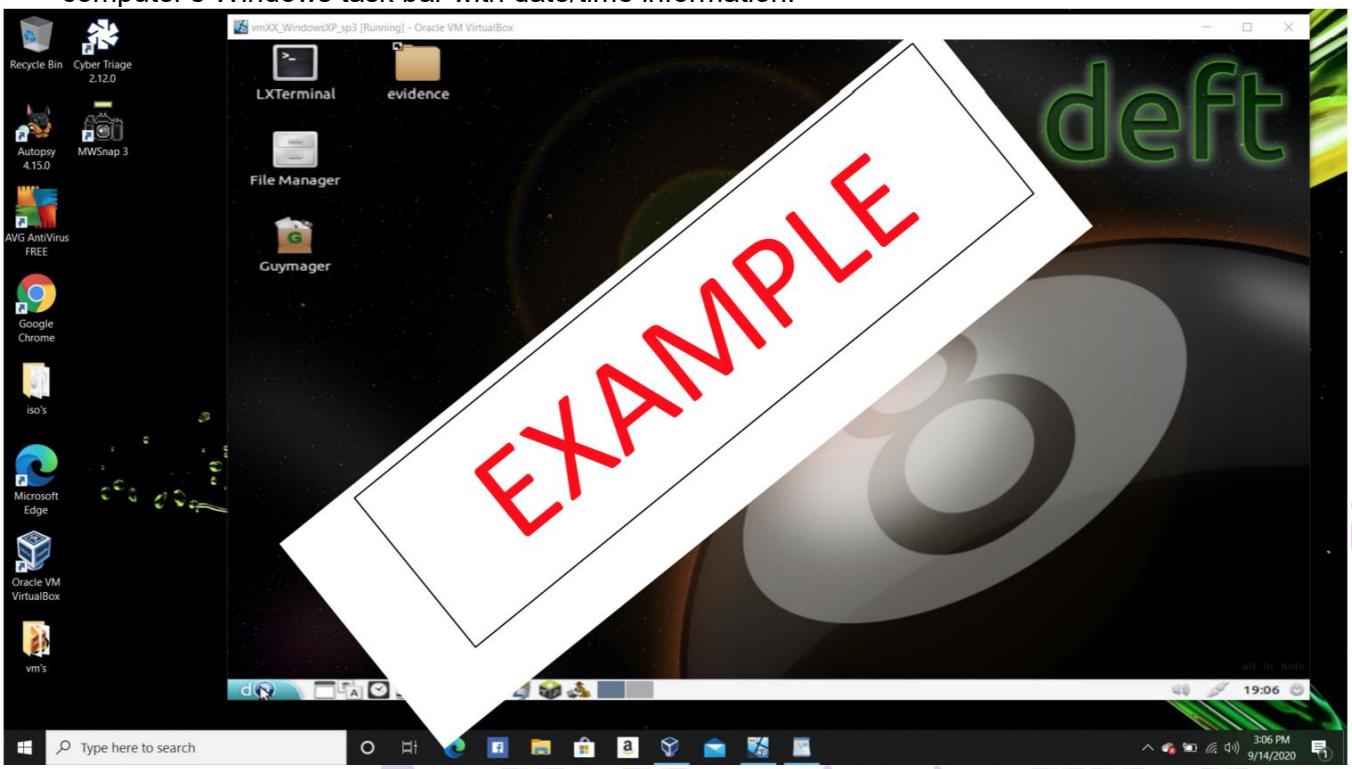
13.6. Example of a scene photo showing the physical evidence computer's BIOS information.



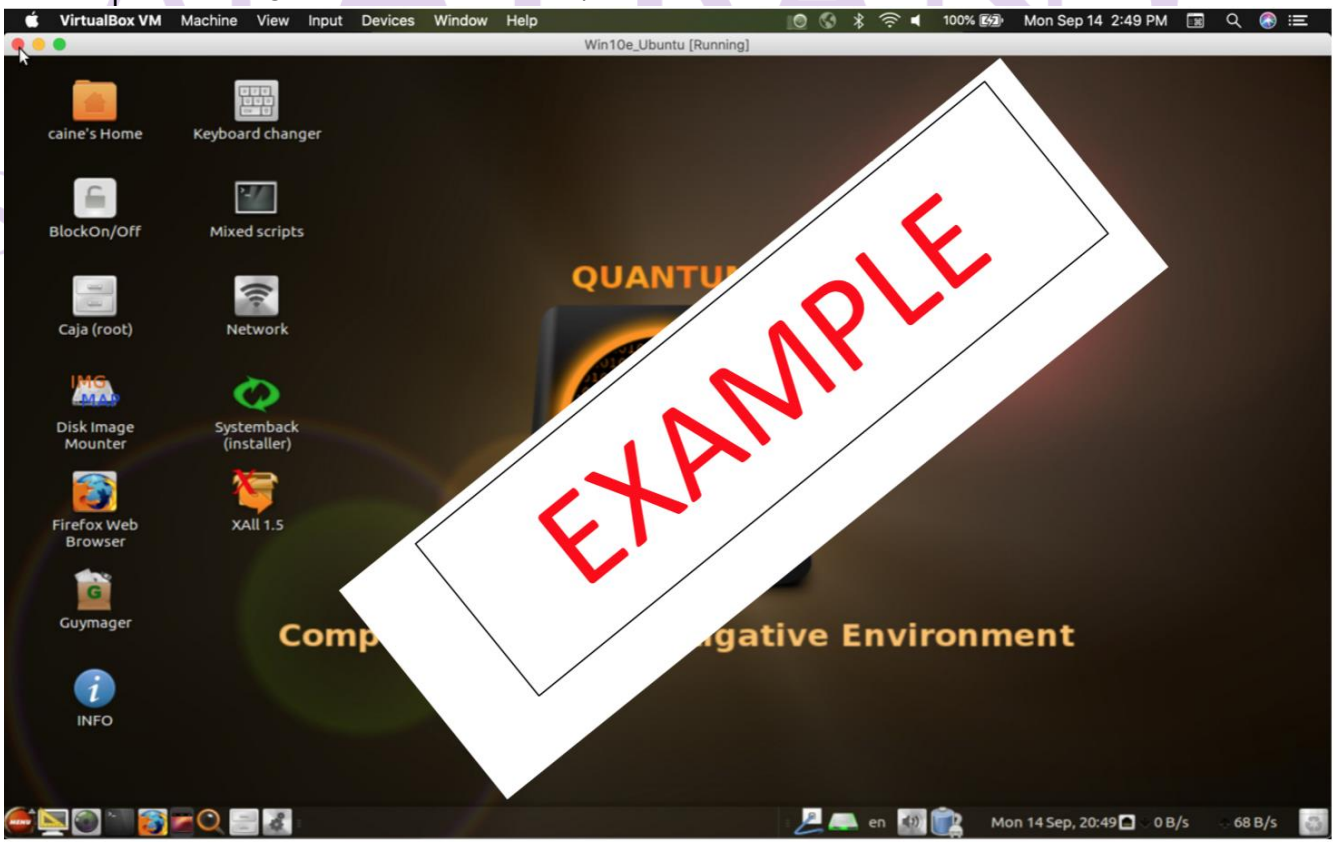

13.7. Example of a virtual scene screenshot showing evidence VM settings via VirtualBox Manager.



13.8. Example of a virtual scene screenshot showing the running evidence VM and evidence computer's Windows task bar with date/time information.




13.9. Example of a virtual scene screenshot showing the live (ON) evidence VM and the evidence computer's MacOSX finder bar with date/time.



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